

Business Impact & Key Insights

- Reduced information retrieval time by replacing manual searches across books, research papers, and websites with instant AI-powered responses.
- Enhanced learning efficiency by translating complex biochemistry concepts into clear, user-friendly explanations.
- Increased user engagement through context-aware, multi-turn conversations that maintain discussion continuity.
- Improved personalization by leveraging MongoDB-based conversation memory, eliminating the need for users to repeat previous queries.
- Delivered a fast, real-time user experience using Groq's low-latency inference, improving responsiveness and usability.
- Unified fragmented knowledge into a single conversational platform, simplifying access to reliable scientific information.
- Designed a scalable and reusable architecture that can be extended to domains such as healthcare, education, legal services, and customer support.
- Demonstrated how Large Language Models (LLMs) combined with retrieval and conversation memory can improve knowledge accessibility while reducing user effort.